

GOVT 6019: Homework 3 — Solutions

Fall 2026 · Probability

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Setup used throughout (the path is relative to this file's folder in the course repository; adjust to your own project's layout):

```
library(tidyverse)
source("../scripts/gov6019_ggplot.R") # theme_gov() and course colors
voter <- read_csv("../data/voter2012.csv")
```

Part A — The rules of probability

Question 1: Two elections

(a) The sample space is the set of registered voters on the county's file: one draw returns one voter, and every voter is an outcome. A^c is the event that the voter did not vote in the spring primary. $A \cap B$ is the event that the voter voted in both elections.

(b) Inclusion–exclusion:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = .24 + .62 - .21 = .65.$$

The subtracted term is the 21 percent who appear in both $\mathbb{P}(A)$ and $\mathbb{P}(B)$ and would otherwise be counted twice.

(c) “Neither” is the complement of “at least one”:

$$\mathbb{P}((A \cup B)^c) = 1 - \mathbb{P}(A \cup B) = 1 - .65 = .35.$$

(d) From (b): “exactly one” is “at least one” minus “both”,

$$\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = .65 - .21 = .44.$$

Separately, $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = .24 - .21 = .03$ and $\mathbb{P}(A^c \cap B) = .62 - .21 = .41$. These two events are disjoint, so additivity gives $.03 + .41 = .44$. The two routes agree.

(e) $A \cap B$ is contained in A , so its probability cannot exceed $\mathbb{P}(A) = .24$. The colleague's .30 does exceed it. The cleanest way to see the violation: A splits into the disjoint pieces $A \cap B$ and $A \cap B^c$, so additivity (axiom 3) forces $\mathbb{P}(A \cap B^c) = .24 - .30 = -.06$. A negative probability contradicts axiom 1. Axioms 1 and 3 together rule the number out.

Question 2: Regimes and leader exits

(a) Summing across each row gives the marginal distribution of regime type: $\mathbb{P}(\text{democracy}) = .002 + .548 = .550$ and $\mathbb{P}(\text{autocracy}) = .018 + .432 = .450$. Summing down each column gives the marginal distribution of leader exit: $\mathbb{P}(\text{irregular}) = .002 + .018 = .020$ and $\mathbb{P}(\text{no irregular exit}) = .548 + .432 = .980$. Each marginal sums to 1.

(b) Conditioning restricts the table to one row and renormalizes:

$$\mathbb{P}(\text{irregular} \mid \text{autocracy}) = \frac{.018}{.450} = .040, \quad \mathbb{P}(\text{irregular} \mid \text{democracy}) = \frac{.002}{.550} = .0036.$$

An autocracy-year carries roughly eleven times the risk of an irregular exit that a democracy-year does.

(c) Conditioning on the column instead:

$$\mathbb{P}(\text{autocracy} \mid \text{irregular}) = \frac{.018}{.020} = .90.$$

(d) Part (b) is a rate: among autocracy-years, the share ending in an irregular exit. Part (c) is a composition: among irregular exits, the share occurring in autocracies. Ninety percent of irregular exits happen under autocracy, and yet any particular autocrat-year ends that way only 4 percent of the time, because autocracy-years are far more numerous than irregular exits. The question “how dangerous is it to be an autocrat?” is answered by (b), the 4 percent. This is the same asymmetry as $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$.

(e) Independence requires $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ in every cell. Take the autocracy/irregular cell:

$$\mathbb{P}(\text{autocracy})\mathbb{P}(\text{irregular}) = .450 \times .020 = .009, \quad \mathbb{P}(\text{autocracy} \cap \text{irregular}) = .018.$$

The joint probability is twice the product of the marginals, so the two variables are not independent. The conditional form says the same thing: $.040 \neq .020$.

Question 3: Two kinds of sampling

(i) **An audit.**

(a) $\mathbb{P}(\text{first has a discrepancy}) = 6/40 = .15$.

(b) The multiplication rule, with the second factor conditional on the first:

$$\mathbb{P}(\text{first two both}) = \frac{6}{40} \times \frac{5}{39} = \frac{30}{1560} = .019.$$

The second factor is $5/39$, not $6/40$: one discrepant precinct and one precinct in total have been removed.

(c) Chaining three conditional factors:

$$\mathbb{P}(\text{none of the 3}) = \frac{34}{40} \times \frac{33}{39} \times \frac{32}{38} = \frac{35,904}{59,280} = .606,$$

so $\mathbb{P}(\text{at least one}) = 1 - .606 = .394$.

(d) With replacement the three draws are independent and each has probability $34/40$ of missing:

$$\mathbb{P}(\text{none of the 3}) = \left(\frac{34}{40}\right)^3 = .614, \quad \mathbb{P}(\text{at least one}) = .386.$$

Sampling **without** replacement is the more likely of the two to catch at least one discrepancy, .394 against .386. The reason is that a clean draw removes a clean precinct from the pool, which raises the share of discrepant precincts facing the next draw. With replacement the pool is restored every time, so the odds never improve.

(ii) Replications.

(e) Independence across teams lets the probabilities multiply:

$$\mathbb{P}(\text{all five succeed}) = .80^5 = .328.$$

(f) The complement of “all five succeed” is “at least one does not”, so

$$1 - .328 = .672.$$

Roughly two times in three, a set of five replications of a real effect contains at least one that fails to reproduce it. A single failed replication is therefore weak evidence against a finding.

(g)

k	$\mathbb{P}(\text{all } k \text{ succeed}) = .80^k$
5	.328
8	.168
10	.107
11	.086
12	.069

The probability crosses .10 between $k = 10$ and $k = 11$, so $k = 11$ is the smallest number of teams for which it falls below .10. Each additional team multiplies the probability by another .80, so the sequence falls geometrically and never reaches zero.

(h) The independence comes from how the five experiments were run. Each team works in a different city, draws its own random sample of households, and runs its own randomization, so conditional on canvassing really raising turnout, one team’s result carries no information about another’s. It is a property of the design, not a claim about how canvassing or politics works.

Concrete features that would break it. All five teams could work from the same canvassing script, in which case a single badly written script pulls every result the same way. All five could run in the same city, so one unusual electorate moves all five estimates together. All five could draw households from the same voter file, so one file’s coverage gaps are common to every sample. Or the teams could read each other’s drafts and adjust their designs, which makes the later experiments conditional on the earlier ones rather than independent of them.

Part B — Bayes' rule

Question 4: Signature verification

(a) The county has observed a flag and wants to know whether the ballot is forged, so the quantity is $\mathbb{P}(F \mid \text{Flag})$:

$$\mathbb{P}(F \mid \text{Flag}) = \frac{\mathbb{P}(\text{Flag} \mid F) \mathbb{P}(F)}{\mathbb{P}(\text{Flag} \mid F) \mathbb{P}(F) + \mathbb{P}(\text{Flag} \mid F^c) \mathbb{P}(F^c)}.$$

The system's advertised accuracy is $\mathbb{P}(\text{Flag} \mid F) = .95$, the probability of a flag given a forgery. That is the reverse conditional, not the answer.

(b) With $\mathbb{P}(F) = .0001$:

$$\mathbb{P}(F \mid \text{Flag}) = \frac{.95 \times .0001}{.95 \times .0001 + .02 \times .9999} = \frac{.000095}{.020093} = .0047.$$

About one flagged ballot in 212 is actually forged.

(c) Of the 200,000 ballots, $200,000 \times .0001 = 20$ are forged and 199,980 are legitimate.

- Forged and flagged: $20 \times .95 = 19$.
- Legitimate and flagged: $199,980 \times .02 = 3,999.6$, call it 4,000.
- Flagged in total: $19 + 3,999.6 = 4,018.6$.

The share of flagged ballots that are forged is $19/4,018.6 = .0047$, which reproduces (b). The two bars of the denominator differ by more than two orders of magnitude, and that gap is the whole answer.

(d) Holding the prevalence and the sensitivity fixed and cutting the false positive rate to .002:

$$\mathbb{P}(F \mid \text{Flag}) = \frac{.95 \times .0001}{.95 \times .0001 + .002 \times .9999} = \frac{.000095}{.0020948} = .045.$$

A tenfold improvement in the false positive rate buys a tenfold improvement in the answer. Even so, more than 95 percent of flagged ballots still belong to legitimate voters, because the prior is doing most of the work.

(e) Under the original settings, roughly 4,000 legitimate voters have their ballots set aside and must be contacted before those ballots count, in order to catch 19 forgeries. The cure process therefore carries essentially all of the cost of the system, and the number of voters it must reach is set by the false positive rate rather than by the amount of fraud. If its capacity is smaller than 4,000, or if some voters cannot be reached in time, the system disenfranchises legitimate voters at a rate far above the rate at which it catches forged ballots.

Part C — Random variables and distributions

Question 5: Three coin flips, one experiment

(a) Writing T for treatment and C for control, in subject order, the sample space has $2^3 = 8$ outcomes:

$$\Omega = \{CCC, TCC, CTC, CCT, TTC, TCT, CTT, TTT\}.$$

The three flips are independent and fair, so each outcome has probability $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

(b) X counts the T 's: $CCC \mapsto 0$; $TCC, CTC, CCT \mapsto 1$; $TTC, TCT, CTT \mapsto 2$; $TTT \mapsto 3$. The probability of a value is the total probability of the outcomes mapping to it:

x	Outcomes	$f(x)$
0	1	1/8
1	3	3/8
2	3	3/8
3	1	1/8

(c) Every value of f lies between 0 and 1. They sum to $\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$. And the probability of a set of values is the sum of f over that set, which is how each entry in the table was built from Ω . All three conditions hold, so f is a PMF.

(d) The CDF accumulates the PMF:

$$F(0) = \frac{1}{8}, \quad F(1) = \frac{4}{8}, \quad F(2) = \frac{7}{8}, \quad F(3) = 1.$$

$F(1.5) = \mathbb{P}(X \leq 1.5) = \frac{4}{8}$: no value of X lies strictly between 1 and 1.5, so nothing is added. $F(-2) = 0$. The CDF is defined at every real number, including numbers X never takes.

(e) There is no appropriate answer. X is discrete: it takes four values, each with strictly positive probability. A density is the object that describes a continuous random variable, for which $\mathbb{P}(X = x) = 0$ at every x and probability is a property of intervals. X is described by the PMF in (b) instead.

(f) $X \sim \text{Binomial}(3, 0.5)$. Three independent trials, two outcomes each, the same success probability on every trial, and X counts the successes. The PMF in (b) matches $\binom{3}{x}(.5)^x(.5)^{3-x}$ term by term, with the coefficients 1, 3, 3, 1.

The independence the Binomial requires comes from the coin. The experimenter ran the randomization, so it is a property of a procedure that was actually carried out, not an assumption about the subjects, who may be as similar to one another as they like.

(g) “All three in the same condition” is the event $\{X = 0\} \cup \{X = 3\}$, and the two are disjoint:

$$\mathbb{P}(X = 0) + \mathbb{P}(X = 3) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8} = .25.$$

One experiment in four assigns every subject to the same condition and so yields no comparison at all, which is why experiments assign a fixed number of units to treatment rather than flipping a coin per unit.

Part D — Applied: education and party in the VOTER data

Question 6: A joint distribution estimated from data

(a) The codebook gives `educ` as 1 No HS, 2 HS grad, 3 Some college, 4 2-year, 5 4-year, 6 Post-grad. A four-year degree or more is therefore $\text{educ} \geq 5$, which puts 5 and 6 on one side of the cut and 1 through 4 on the other. The 2-year degree sits with “No BA”: it is a completed credential, but it is not the four-year degree the cut is about.

```
cc <- voter |>
  filter(!is.na(educ), !is.na(pid3)) |>
  mutate(
    ba_f = factor(if_else(educ >= 5, 1, 0), levels = 0:1,
                  labels = c("No BA", "BA or higher")),
    pid3_f = factor(pid3, levels = 1:3,
                  labels = c("Democrat", "Republican", "Other"))
  )

nrow(cc) # complete cases
```

```
[1] 7895
```

```
nrow(voter) - nrow(cc) # rows dropped
```

```
[1] 105
```

```
sum(is.na(voter$educ)) # missing educ
```

```
[1] 17
```

```
sum(is.na(voter$pid3)) # missing pid3
```

```
[1] 89
```

```
sum(is.na(voter$educ) & is.na(voter$pid3)) # missing both
```

```
[1] 1
```

The complete-case table holds 7,895 respondents, and 105 of the 8,000 rows are dropped. Of those, 17 are missing `educ` and 89 are missing `pid3`, with 1 respondent missing both. The counts reconcile as $17 + 89 - 1 = 105$: adding the two without subtracting the overlap would double-count that respondent, which is inclusion–exclusion applied to a pair of missingness events.

Dropping them treats the 105 as exchangeable with the respondents who remain — that is, it assumes the education-by-party distribution among the people who skipped one of the two questions is the same as among those who answered both. The assumption is not testable with this file, and it is not obviously true: declining to state a party is itself political behavior.

(b)

```
joint <- cc |>
  count(ba_f, pid3_f) |>
  mutate(p = n / sum(n)) |>
  select(-n) |>
```

```

pivot_wider(names_from = pid3_f, values_from = p)

joint |>
  mutate(Total = Democrat + Republican + Other) |>
  bind_rows(summarize(joint, ba_f = factor("Total"),
    across(where(is.numeric), sum)) |>
    mutate(Total = Democrat + Republican + Other)) |>
  mutate(across(where(is.numeric), \(x) round(x, 4)))

```

A tibble: 3 x 5

	ba_f	Democrat	Republican	Other	Total
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	No BA	0.27	0.238	0.0931	0.602
2	BA or higher	0.19	0.152	0.056	0.398
3	Total	0.46	0.391	0.149	1

	Democrat	Republican	Other	Total
No BA	.2700	.2385	.0931	.6016
BA or higher	.1900	.1524	.0560	.3984
Total	.4600	.3909	.1491	1.0000

The six cells sum to 1 by construction: each is a count divided by the same total, and the six counts exhaust the complete cases.

(c) $\mathbb{P}(\text{Democrat}) = .2700 + .1900 = .4600$, the column total, summing the two cells in the Democrat column. $\mathbb{P}(\text{BA or higher}) = .1900 + .1524 + .0560 = .3984$, the row total, summing the three cells in the BA row. Each is a marginal: the joint distribution summed over the other variable.

(d)

```

tab <- table(cc$ba_f, cc$pid3_f)
round(prop.table(tab, 1), 4)      # party given education

```

	Democrat	Republican	Other
No BA	0.4488	0.3964	0.1547
BA or higher	0.4769	0.3825	0.1405

```

round(prop.table(tab, 2), 4)      # education given party

```

	Democrat	Republican	Other
No BA	0.5870	0.6102	0.6245
BA or higher	0.4130	0.3898	0.3755

$$\mathbb{P}(\text{Dem} \mid \text{BA or higher}) = \frac{.1900}{.3984} = .477, \quad \mathbb{P}(\text{Dem} \mid \text{No BA}) = \frac{.2700}{.6016} = .449,$$

$$\mathbb{P}(\text{BA or higher} \mid \text{Dem}) = \frac{.1900}{.4600} = .413.$$

The first two condition on a row and divide by that row's total; the third conditions on a column and divides by that column's total.

(e)

```
pe <- rowSums(prop.table(tab))      # marginal: education
pp <- colSums(prop.table(tab))      # marginal: party
round(prop.table(tab) - outer(pe, pp), 4)  # joint minus product of marginals
```

	Democrat	Republican	Other
No BA	-0.0067	0.0033	0.0034
BA or higher	0.0067	-0.0033	-0.0034

```
max(abs(prop.table(tab) - outer(pe, pp)))
```

```
[1] 0.006735971
```

Cell by cell, the joint relative frequencies and the products of the marginals are:

	Democrat	Republican	Other
No BA, joint	.2700	.2385	.0931
No BA, product	.2768	.2352	.0897
BA or higher, joint	.1900	.1524	.0560
BA or higher, product	.1833	.1557	.0594

Independence requires equality in every cell, and it fails in all six. The largest discrepancy is .0067, in the two cells of the Democrat column.

Over-represented relative to independence: BA/Democrat and No BA/Republican, together with No BA/Other. Under-represented: No BA/Democrat, BA/Republican and BA/Other. The pattern is one gradient rather than six unrelated gaps. Degree holders are more Democratic and less Republican than independence would put them, and respondents without a four-year degree are the reverse. The conditional distributions in (d) say the same thing in a single number: $\mathbb{P}(\text{Dem})$ rises from .449 to .477 across the cut, a gap of 2.8 percentage points.

(f) For the statement to be about the American electorate rather than about these respondents, the 8,000 would have to be a random sample of that electorate, with nonresponse unrelated to education and party. Neither condition is guaranteed here: the sample is an online panel, and the 105 respondents dropped in (a) were dropped on the very two variables at issue.

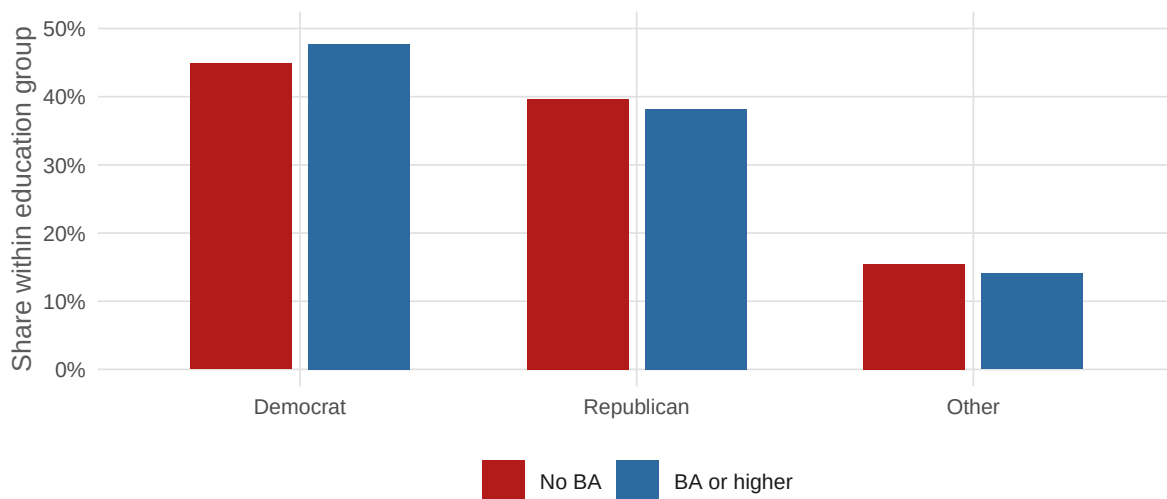
To decide whether .0067 is more than an artifact of which 8,000 people were sampled, you would need a measure of how much a discrepancy like this varies from one sample to the next when the two variables really are independent. The observed number alone cannot be assessed, since some discrepancy will appear in any finite sample whatever the population looks like.

(g)

```
cc |>
  count(ba_f, pid3_f) |>
  group_by(ba_f) |>
  mutate(share = n / sum(n)) |>
  ungroup() |>
```

```
ggplot(aes(x = pid3_f, y = share, fill = ba_f)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_fill_manual(values = unname(gov_colors[c("carnelian", "blue")] ),
    name = NULL) +
  scale_y_continuous(limits = c(0, 0.5),
    labels = scales::label_percent(accuracy = 1)) +
  labs(
    title = "Degree holders lean more Democratic, by under three points",
    x = NULL, y = "Share within education group"
  ) +
  theme_gov(base_size = 10)
```

Degree holders lean more Democratic, by under three points



Respondents with a four-year degree identify as Democrats 2.8 percentage points more often than respondents without one, and as Republicans 1.4 points less often. The figure describes 7,895 survey respondents in a single year, and a binary education cut hides whatever happens between the no-high-school and post-graduate ends of the original six-category scale.

Shares within education group, not counts: the two groups differ in size — 60 percent against 40 percent — so raw counts would not be comparable across them.